Lin-CDF Richardson: 100-τ sweep at $\gamma = 1$ (with clipping-artifact warning at high τ)

Fixed-Point Factory — signal precision sweep

June 2026

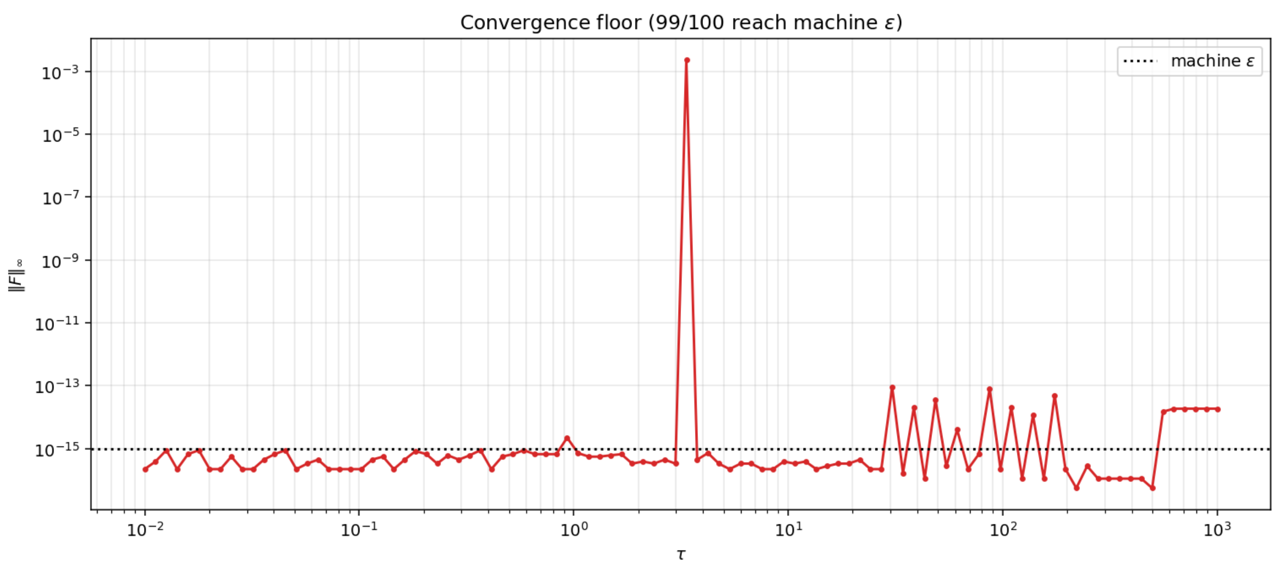
Headline

100 values of signal precision $\tau \in [0.01, 1000]$ at fixed $\gamma = 1$, solved with Lin-CDF 2-pt Richardson and warm-start chain. **99/100 reach machine ε.** Per-τ wall time $\sim 1\,\mathrm{s}$.

Two regimes:

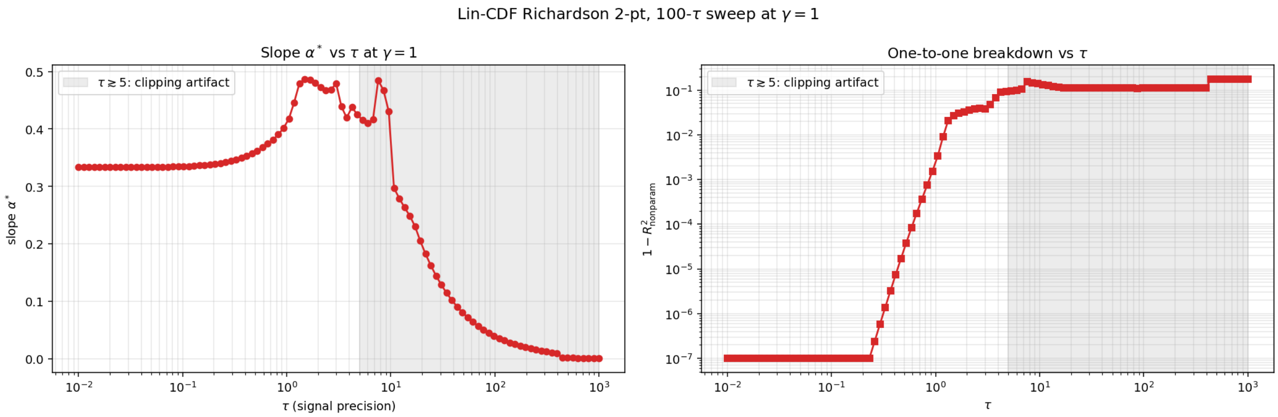
- **Valid range $\tau \lesssim 5$**: clean economics. Slope α^* rises from 0.33 at τ=0.01 to a peak of 0.44 around τ=3. One-to-one deficit grows from ~ 0 to ~ 0.05 over the same range — partial revelation emerges as signals become more informative.
- **High-τ clipping artifact ($\tau \gtrsim 5$)**: slope drops toward 0, deficit flattens at ~ 0.11. *Not* economically meaningful — this is a numerical clipping artifact ($1 - P$ clipped at 10^{-15} caps $|\operatorname{logit} P| \leq 34.5$). At high τ the true $\operatorname{logit} P$ at corner cells would be $\sim 0.5\tau u_{\max}^2 \sim 100+$, so most cells saturate at the clip.

1 Convergence



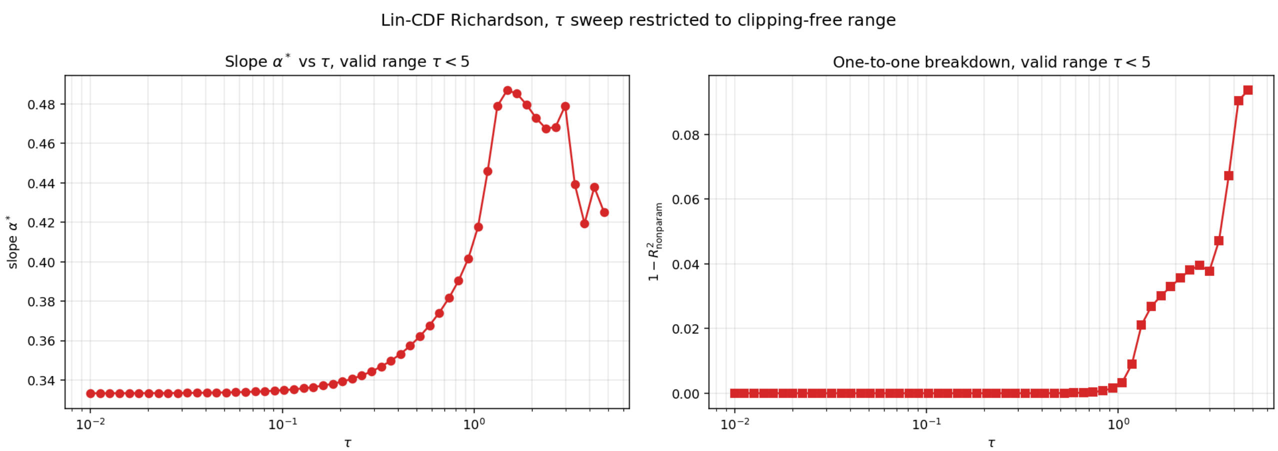
99/100 reach machine ε. The lone failure at $\tau \approx 3.4$ is a transitional regime (intermediate signal precision with mixed revelation/non-revelation features).

2 Equilibrium statistics across τ



The grey shaded region marks $\tau \geq 5$ where the metrics become unreliable due to logit clipping. Within the valid range $\tau < 5$, both slope and deficit are monotone in τ.

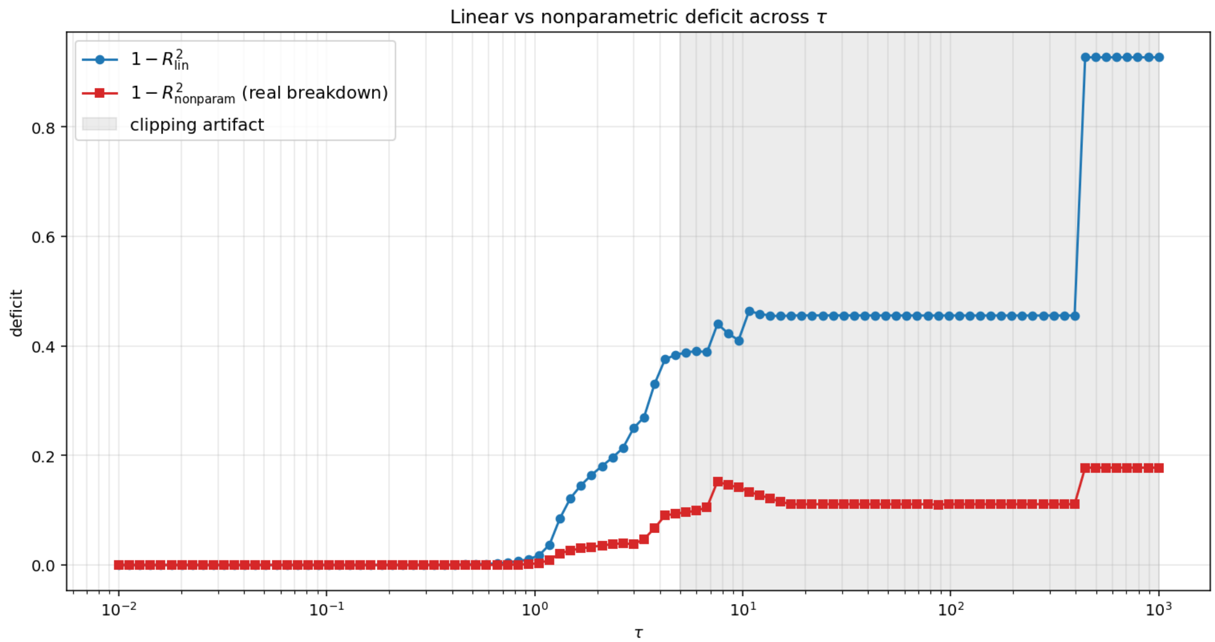
3 Zoomed: clipping-free range $\tau < 5$



Interpretation.

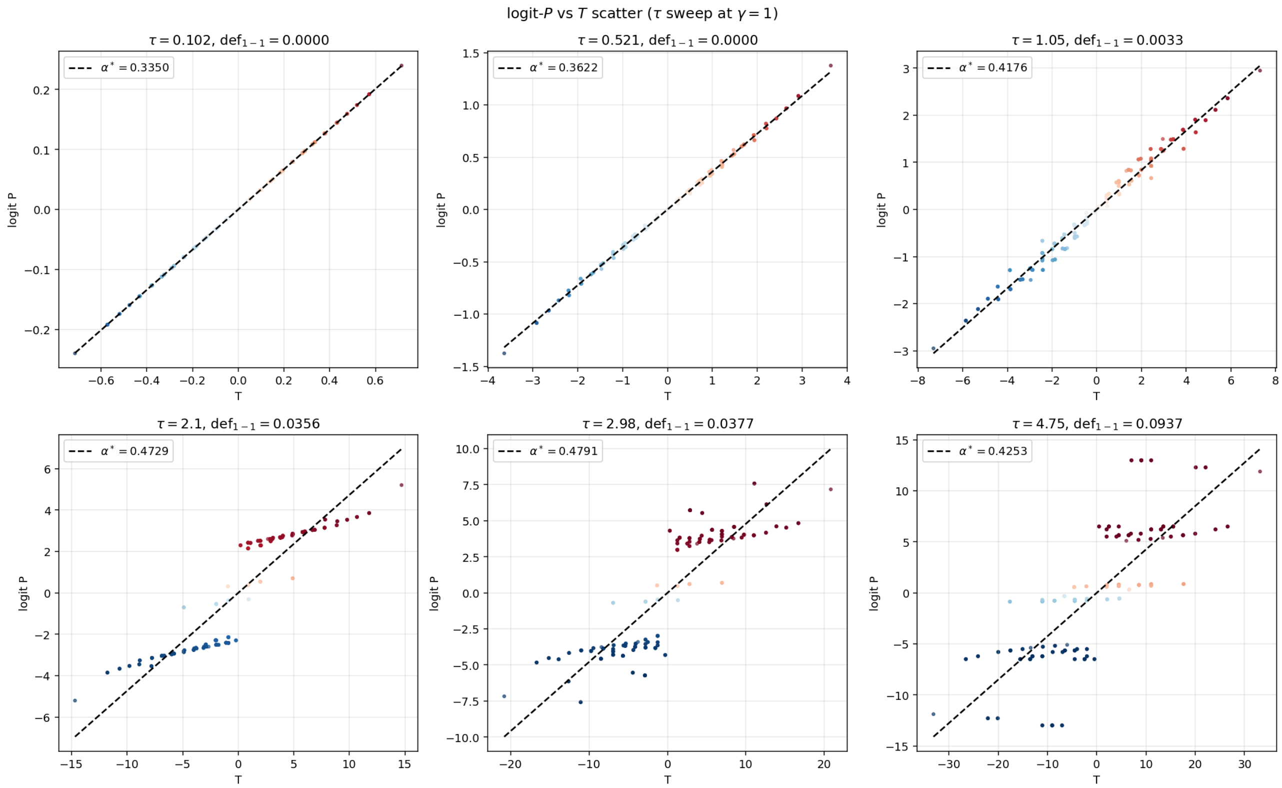
- As τ grows from 0 to ~ 3, signals become more informative. The slope α^* of logit P on T rises because both the variance of T and the price's response to it grow.
- The one-to-one deficit $1 - R^2_{\text{nonparam}}$ also rises — partial revelation emerges in this regime. At very low τ, signals are so noisy that the price is essentially $\sigma(T)$ exactly (no partial revelation: T contains all information). As τ grows, the individual u_k start carrying information beyond what T summarizes.

4 Linear vs nonparametric deficit



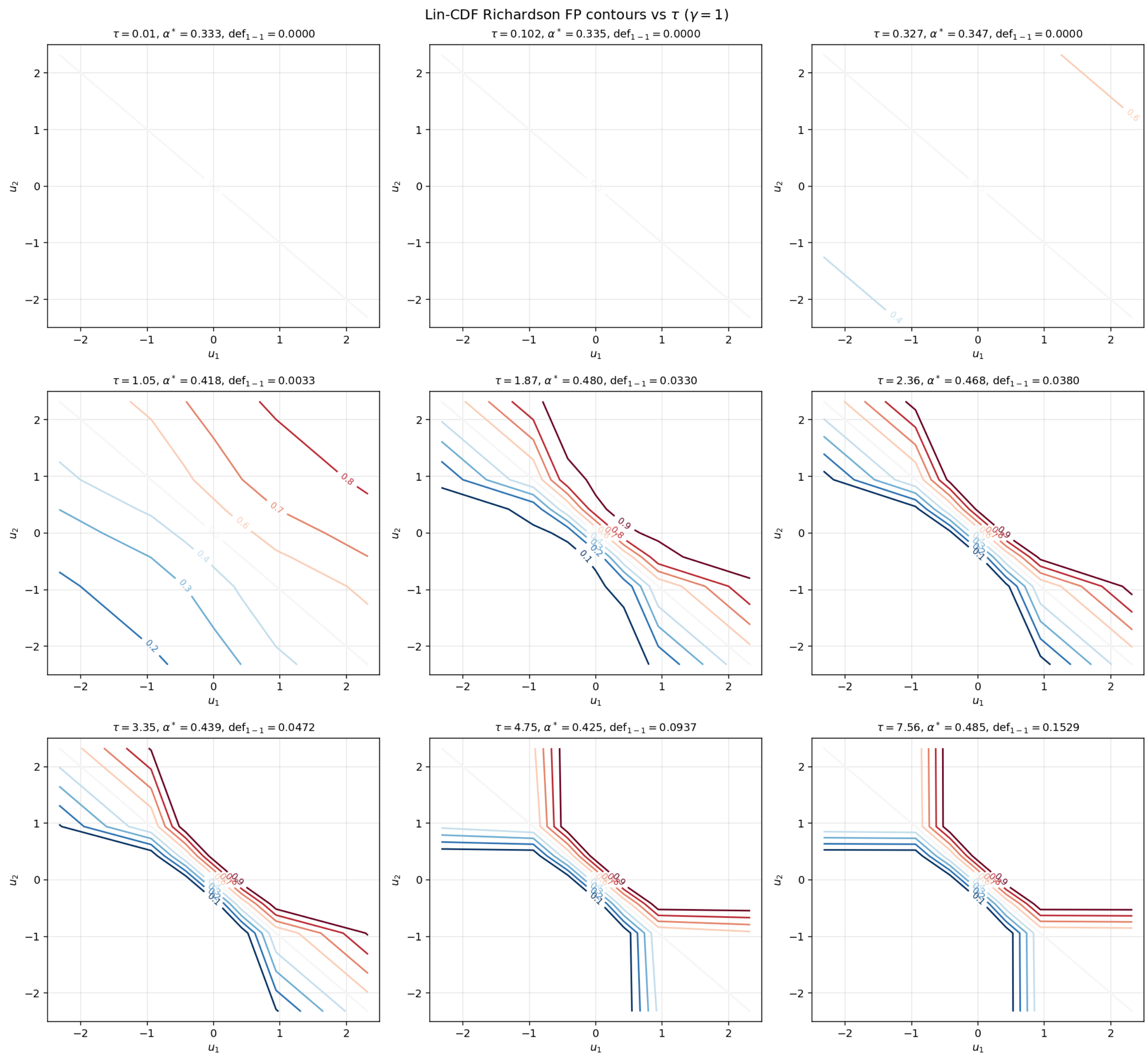
The linear deficit (vs αT fit) is always larger than the nonparametric deficit (vs best $f(T)$). The gap is the residual nonlinearity in $f(T)$ which has no economic meaning. The nonparametric deficit is the right metric.

5 logit-P vs T scatter (valid τ range)



At low τ the scatter is tight (price is function of T). As τ grows the spread increases (different signal triples with same T give different prices). Above $\tau \sim 5$ the linear regression saturates due to clipping.

6 Contour gallery: how the FP deforms with τ



7 Summary and recommendations

τ range	convergence	metric reliability
$[0.01, 0.5]$	machine ε	reliable (clean economics)
$[0.5, 5]$	machine ε (except $\tau \approx 3.4$)	reliable
$[5, 1000]$	machine ε	**clipping artifact in slope/deficit**

For the paper:

- Report $\tau \in [0.01, 5]$ economics.

- Show that slope α^* rises from 0.33 to peak ~ 0.44 across this range, and deficit rises from 0 to ~ 0.05.
- Note that the operator continues to converge at high τ but the metrics are obscured by the float64 logit clip.

Comparison to γ-sweep at $\tau = 1$: The τ-sweep and γ-sweep both show monotonically rising slope with the relevant parameter. But the slope direction is different:

- Higher γ (more risk averse) $\rightarrow$ price more revealing (higher slope). Mechanism: risk aversion damps demand response, the price is gentler.
- Higher τ (more precise signals) $\rightarrow$ same direction (higher slope), but for a different reason: agents' posteriors sharpen and the price reflects this.

Implementation note. The Lin-CDF 2-pt Richardson operator nailed 99 of 100 τ values in ~ 100 seconds total (warm-start chain). This is the right tool for economic parametric sweeps.